

Theorem 2: Gamma-Normalization and Fractional Pi

algebraicity of normalized periods and the radical constants

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The «Dynamic Principle» Program — the hodge-laboratory

Standalone theorem monograph · Монография 5/16

1 Setting

The Γ -normalization absorbs the transcendental factor of every period, turning normalized periods into algebraic numbers of the phase lattice $\frac{1}{N}\mathbb{Z}[\zeta_N]$. The program's requirement — fractional shares of π instead of $4\pi^2$ -normalizations — holds structurally on the stand.

Definition 1 (Γ -normalization). For $P \in \frac{\Omega_{a,b}}{N}\mathbb{Z}[\zeta_N]$ the normalized period is $\hat{P} = P/\Omega_{a,b} \in \frac{1}{N}\mathbb{Z}[\zeta_N]$.

Theorem 2 (Γ -normalization and fractional π). Let $N \in \{15, 30\}$. Then: (1) algebraicity: $\hat{P} \in \frac{1}{N}\mathbb{Z}[\zeta_N]$ — the phase lattice has denominator N ; (2) reduction: each $\Omega_{a,b}$ is expressed through canonical Γ -values and factors $\pi/\sin(\pi k/N)$; at $a+b = N$ — a pure share of π ; (3) no $4\pi^2$: π enters only via $\pi/\sin(\pi k/N)$ and the multiplication formula; no power $4\pi^2$ arises in the normalization; (4) radicals:

$$b_{Ch}(15) = 1 - \frac{1 + \sqrt{5} + \sqrt{3} \sqrt{10 - 2\sqrt{5}}}{8}, \quad (1)$$

$$b_{Ch}(30) = 1 - \frac{\sqrt{30 + 6\sqrt{5}} + \sqrt{5} - 1}{8}. \quad (2)$$

Proof. (1) — dividing the lattice $\frac{\Omega}{N}\mathbb{Z}[\zeta_N]$ by Ω . (2)–(3) — the inventory: reflection introduces π only as $\pi/\sin(\pi k/N)$, multiplication only as $(2\pi)^{(m-1)/2}$ at integer m ; the normalization is built from the periods themselves, so $4\pi^2$ is not generated. (4): $\cos 24^\circ = \cos(60^\circ - 36^\circ)$ with $\cos 36^\circ = \frac{1+\sqrt{5}}{4}$, $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$ gives the level-15 radical; $\cos 12^\circ = \cos(30^\circ - 18^\circ)$ with $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$, $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ gives the level-30 one. Numerically: 0.086454 and 0.021854. $\square$

Remark 3. The share of π is always rational with denominator N : the factor $\pi/\sin(\pi k/N)$ carries the angle $k\pi/N$; the phase lattice carries the denominator N in algebra. A structural claim — true for all blocks at once. The ladder $\pi/7, \pi/15, \pi/30$ is one μ_N -quantum ladder (certificates G, H).

Protocol data: numbers

$b_{Ch}(7) \approx 0.7526$; $b_{Ch}(9) \approx 0.1172$; $b_{Ch}(11) \approx 0.0823$; $b_{Ch}(15) \approx 0.086454$; $b_{Ch}(30) \approx 0.021854$. The universal Tate twist $4\pi^2$ lives in the lattice spectra (G1), not in the period normalization.

Summary of results: summary

The theorem fixes: the phase-lattice denominator equals the level N ; the transcendental part reduces to fractional π ; the radical constants are written in closed form.

Verification: reproduction

`python3 laboratory.py` → “Stand N=15/30”; Lean: `bch_radicals`; the plot `fig_bch.png`.

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